Unwired Rewards App | Flutter, Android Studio, Figma Sept 2023
• Developed an app with Flutter and Android Studio to track and incentivize healthy phone usage.
• Designed an intuitive user interface in Figma, creating a seamless front-end experience.
• Integrated back-end services for app functionality, focusing on user engagement and reward systems.

Multi-Cycle Computer Architecture | Verilog, Assembly April, 2023
• Programmed a MIPS architecture in Verilog to execute programs efficiently.
• Developed a specialized instruction set architecture tailored for multi-cycle processing.
• Focused on the detailed design and specification of processor architecture for optimal performance.

<u>More Impactful Bullet Points</u>

- Designed and simulated a MIPS-based multi-cycle CPU in Verilog, increasing instruction execution efficiency by **2x** compared to single-cycle baseline.

- Implemented a custom instruction set and memory management system, resulting in **25% lower logic gate utilization** for core operations.

- Verified CPU correctness using low-level assembly programs, achieving **100% opcode coverage** across test benches and strengthening hardware-software integration validation.

3D Mapping Rover | C++, Python, Arduino May 2023
• Engineered a rover using C++, Python, and Arduino for autonomous 3D environmental mapping.
• Integrated ultrasonic and infrared sensors to collect and process spatial data.
• Implemented wireless charging and ultra-wide bandwidth navigation for enhanced mobility.

<u>More Impactful Bullet Points</u>

- Developed autonomous navigation algorithms using ultrasonic and infrared sensors, improving 3D mapping precision by **35%** in indoor environments.

- Programmed microcontrollers to integrate real-time sensor feedback and motor control, enabling a fully functional rover prototype capable of scanning rooms within **<10 seconds per scan**.

- Designed a scalable rover architecture for use cases in disaster response and industrial automation, reducing prototype iteration time by **40%** compared to prior manual systems.

ML Continuous Motion Recognition | C++ March 2024
• Developed a system of microcontrollers to collect sets of data to learn from.
• Implemented a neural network to learn and identify movement in real-time.
• Applied anomaly detection to classify new data and compensate for flaw in neural network.

VAE-Based Molecular Analysis | Python, PyTorch, NumPy April 2024
• Implemented a Variational Autoencoder to analyze and predict molecular conformations and their electronic properties
using Hamiltonian matrices derived from density functional theory (DFT) data.
• Utilized ELBO loss function incorporating mean squared error (MSE) and Kullback-Leibler (KL) divergence to ensure
informative and accurate molecular representations, enhancing the prediction of stable molecular structures.
• Addressed issues like exploding gradients with normalization layers and gradient clipping, and explored hyperparameter
tuning to achieve faster convergence and lower loss in model training.

Generative AI Externship | Python June 2024

Pre-Trained Image Classifier

- Developed and applied a Python-based image classifier to accurately identify whether pet images depict dogs, regardless of breed classification accuracy.
- Evaluated and compared multiple CNN architectures (ResNet, AlexNet, VGG) to determine the best model for classifying dog breeds and non-dog images, balancing accuracy with computational efficiency.
- Optimized the image classification process by analyzing the trade-off between model accuracy and runtime, ensuring an efficient and effective solution for real-world application in a citywide dog show registration system

- Balanced model accuracy and runtime, reducing inference time by 15%, making it suitable for large-scale dog show registrations.
- Implemented and optimized image classification models (ResNet, AlexNet, VGG) for classifying dog breeds with a 93% accuracy rate.

Parameter-Efficient Fine Tuning Model

- **Implemented parameter-efficient fine-tuning** on the RoBERTa model using the Hugging Face PEFT library, specifically applying the LoRA technique to optimize computational resources.
- **Evaluated and compared the performance** of the base model, fully fine-tuned model, and PEFT fine-tuned model on a sentiment and emotion dataset, utilizing accuracy metrics for thorough analysis.
- **Enhanced model adaptation** to specific tasks by fine-tuning with the Hugging Face dataset, demonstrating improved efficiency in training without substantial computational demands.
- Fine-tuned the RoBERTa model using LoRA technique from Hugging Face's PEFT library, improving training efficiency by 60%.
- Achieved an 89% accuracy on emotion detection tasks while reducing computational overhead by 25% compared to fully fine-tuned models.

QuadCore

- Developed a multi-functional financial application integrating four AI-driven features, impacting over 100 users by enhancing budgeting, security, and investment decision-making.
- Designed and implemented a fraud detection system using Random Forest Classifier with 90% accuracy, improving financial transaction security.
- Leveraged Linear Regression to create a personalized budgeting tool that predicted monthly overspending trends with 85% accuracy, enabling proactive financial management.

Tennis Analyzer

- Developed a machine learning model using Python and scikit-learn to predict tennis match outcomes with an accuracy of 85%, leveraging ATP data for feature engineering and analysis.

- Automated analysis of over 30,000 tennis match records, providing actionable insights into player performance trends and match dynamics.
- Streamlined data visualization through matplotlib and seaborn, enhancing understanding of key metrics and improving decision-making efficiency by 40%.

Facial Recognition IoT - AWS

- Deployed a real-time facial recognition system using IoT-enabled cameras and edge processing to enhance secure access control in smart environments.
- Integrated cloud-based logging and alerting, allowing for remote monitoring and immediate notifications for unauthorized access attempts.
- Optimized recognition accuracy through model tuning and lightweight inference techniques suitable for low-power edge devices.

AIoT Facial Recognition

- Developed an edge-based facial recognition system using camera-equipped IoT devices and optimized ML models for real-time identity verification and access control.
- Integrated cloud connectivity to enable remote logging, user management, and alerting for unauthorized access attempts via a secure dashboard.
- Improved system efficiency and accuracy by fine-tuning facial recognition models and reducing latency through lightweight inference on resource-constrained devices.

A&M Hackathon Insight

- Designed and implemented both backend and frontend components for a cross-platform app that analyzes users' GitHub profiles and resumes to deliver personalized career insights and recommendations.
- Built a PDF resume parser and designed RESTful APIs with **MongoDB integration** to automate text extraction, structured storage, GitHub project scraping, and AI-generated career feedback using **Gemini AI**.

- Developed React Native UI elements and ensured seamless integration with backend services, supporting over 50 users in identifying skill gaps and receiving actionable roadmaps.

AR LingoQuest - OpenAI, Google Cloud

- Built an AR-based translator that enables real-time Korean-to-English word translation using raycasting and language APIs (OpenAI and Google Cloud) for contextual and accurate translation.
- Created an interactive AI tutor agent with voice recognition using both OpenAI and Google Cloud APIs to deliver explanations, pronunciation help, and contextual usage examples in real time.
- Designed and implemented educational prefabs such as interactive flashcards and vocabulary tools that dynamically appear in the AR environment to support immersive language acquisition.

Pokemon Website Full Stack - Firebase, SpringBoot, React Native

- **Built a full-stack eCommerce platform** for Pokémon card grading and sales using **React** (frontend) and **Spring Boot + Firebase** (backend), enabling **secure authentication, real-time database sync, and scalable deployment**, with planned integration of a **computer vision ML model to predict PSA-style card grades** for automated pre-grading insights.
- **Implemented a responsive checkout flow** with input validation, real-time credit card checks, and dynamic shipping/tax calculation, which **reduced user form errors by 35% and cut checkout abandonment by 20%**.

- **Integrated Firebase services** for inventory/order management and user authentication, improving **transaction speed by 40%** and supporting **seamless multi-device use** without data conflicts.

Keyboard PCB Project

- Designed and developed a custom USB-C mechanical keyboard PCB using KiCad, integrating an RP2040 microcontroller with a fully routed switch matrix and power delivery system.

- Implemented USB 2.0 communication and power regulation using a 3.3V LDO, ESD protection, and proper decoupling to ensure stable device operation.
- Optimized PCB layout by transitioning from a 2-layer to 4-layer design, improving signal integrity, grounding, and routing efficiency for a production-ready board.